Definition 1. Let Var be the set of propositional variables. Then ϕ is defined inductively as follows:

$$\phi ::= p \mid \phi \vee \phi \mid X\phi \mid \phi U \psi \mid \langle a \rangle \phi$$

Definition 2. Let ϕ be a formula and $p \in Var$. Then the Closure of ϕ is inductively defined as

$$\begin{aligned}
Cl(p) &= Cl(\neg p) = \{p, \neg p\} \\
Cl(\neg\phi) &= Cl(\phi) \\
Cl(\phi \vee \psi) &= \{\phi \vee \psi, \neg(\phi \vee \psi)\} \cup Cl(\phi) \cup Cl(\psi) \\
Cl(X\phi) &= \{X\phi, \neg X\phi\} \cup Cl(\phi) \\
Cl(\phi U \psi) &= \{\phi U \psi, \neg(\phi U \psi), trueU\langle a\rangle\psi, \neg(trueU\langle a\rangle\psi)\} \cup Cl(\phi) \cup Cl(\psi) \\
Cl(\langle a\rangle\phi) &= \{\langle a\rangle\phi, \neg\langle a\rangle\phi\} \cup Cl(\phi)
\end{aligned}$$

Definition 3. An atom A is a subset of $Cl(\phi)$ such that it is consistent that means $A \nvdash \bot$ and

$$\begin{aligned}
&\text{if } \psi \in Cl(\phi), && \text{then } \psi \in A \text{ iff } \neg\psi \notin A, \\
&\text{if } \phi_1 \vee \phi_2 \in Cl(\phi), && \text{then } \phi_1 \vee \phi_2 \in A \text{ iff } \phi_1 \in A \text{ or } \phi_2 \in A \\
&\text{if } \phi_1 U \phi_2 \in Cl(\phi), && \text{then if } \phi_1 U \phi_2 \in A \text{ then } \psi_1 \in A \text{ or } \psi_2 \in A \\
&\text{if } \phi_1 U \phi_2 \in Cl(\phi), && \text{then if } \phi_2 \in A \text{ then } \phi_1 U \phi_2 \in A
\end{aligned}$$

Lemma 4. *Assume $A_i \vdash \psi$ For some $\psi \in Cl(\phi)$. Then $\psi \in A_i$.*

Proof. Assume $\psi \notin A_i$. That means $\psi \in A_i$. But then $A_i \vdash \psi$ and $A_i \vdash \neg\psi$ which is a contradiction. □

Definition 5. We say two atoms A, B are in relation $R_x (AR_xB)$ iff

$$\begin{aligned}
&\text{if } X\psi \in Cl(\phi), && \text{then } X\psi \in A \text{ iff } \psi \in B, \\
&\text{if } \psi_1 U \psi_2 \in Cl(\phi), && \text{then if } \psi_1 U \psi_2, \neg\psi_2 \in A, \text{ then } \psi_1 U \psi_2 \in B, \\
& && \text{and if } \psi_1 U \psi_2 \in B, \psi_1 \in A, \text{ then } \psi_1 U \psi_2 \in A.
\end{aligned}$$

Lemma 6. *For any atom A there exists an atom B s.t. AR_xB.*

Proof. Assume A is an atom. Then we define B' as follows:

$$B' = \{\phi \mid X\phi \in A\} \cup \{\neg\phi \mid \neg X\phi \in A\} \cup \{\psi_1 U \psi_2 \mid \psi_1 U \psi_2, \neg\psi_2 \in A\} \cup \{\neg(\psi_1 U \psi_2) \mid \neg(\psi_1 U \psi_2), \psi_1 \in A\}$$

Now we show that B is consistent. Assume it is not. Then there exists $\alpha_1, ..., \alpha_n$ s.t. $B \vdash \alpha_1 \wedge \alpha_2 ... \wedge \alpha_n \rightarrow \bot$. Let's define $\hat{B} = \bigwedge B$. It follows:

$$
\begin{array}{ll}
\vdash \hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n \rightarrow \bot & \\
\vdash X(\hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n \rightarrow \bot) & \text{By X-necessitation} \\
\vdash X(\hat{B} \wedge \alpha_1 \wedge ... \wedge \alpha_n) \rightarrow X\bot & \text{F2} \\
\vdash X(\hat{B}) \wedge X\alpha_1 \wedge ... \wedge X\alpha_n \rightarrow X\bot &
\end{array}
$$

Assume, $a_i = \phi$. Then by definition we have $A \vdash Xa_i$. If $a_i = \neg\phi$, then we can use F1 to obtain $A \vdash X\neg\phi$. If $a_i = \psi_1 U \psi_2$ then by definition, $\psi_1 U \psi_2 \in A$ and $\neg\psi_2 \in A$. Consider the following proof steps:

$$
\begin{array}{ll}
A \vdash \neg\psi_2 & \\
A \vdash \psi_1 U \psi_2 & \\
A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 & \\
A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \neg\psi_2 & \text{Unfolding U} \\
A \vdash \psi_1 U \psi_2 \wedge \neg\psi_2 \rightarrow (\psi_1 \wedge X(\psi_1 U \psi_2)) & \text{distributivity and other rules} \\
A \vdash \psi_1 \wedge X(\psi_1 U \psi_2) & \\
A \vdash X(\psi_1 U \psi_2). &
\end{array}
$$

Hence, $A \vdash Xa_i$. If $a_i = \neg(\psi_1 U \psi_2)$, then by definition, $\neg(\psi_1 U \psi_2), \psi_1 \in A$. Our goal is to derive $A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1 \rightarrow X\neg(\psi_1 U \psi_2)$.

$$
\begin{array}{ll}
A \vdash \neg(\psi_1 U \psi_2) \wedge \psi_1 & \\
A \vdash \neg(\psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2))) \wedge \psi_1 & \text{Unfolding U} \\
A \vdash \neg\psi_2 \wedge (\neg\psi_1 \vee \neg X(\psi_1 U \psi_2)) \wedge \psi_1 & \\
A \vdash \neg\psi_1 \vee \neg X(\psi_1 U \psi_2) & \\
\neg X(\psi_1 U \psi_2) & \psi_1 \text{ is derivable} \\
A \vdash X\neg(\psi_1 U \psi_2) & \text{F1}
\end{array}
$$

By similar argument, we can conclude $A \vdash X\hat{B}$. So it follows

$$A \vdash X\bot$$

Because $X\bot \rightarrow \bot$ is a tautology, we conclude $A \vdash \bot$ which is a contradiction. Now, by the Extension Lemma, we can extend B' to an atom B. It is easy to check, that AR_xB.

□

Lemma 7. *Let $\Gamma = \{\phi\}'$ be a consistent set of formulas. Then Γ' can be extended to an atom.*

Proof. Let $n = |CL(\phi)|$ and consider the enumeration $\psi_1, \psi_2, ..., \psi_n$ of $CL(\phi)$. We define $\Gamma_0 = \{\phi\}$ and then inductively

$$\Gamma_{n+1} := \begin{cases} \Gamma_n \cup \psi_{n+1}, & \text{if } \Gamma_n \cup \psi_{n+1} \text{ is consistent,} \\ \Gamma_n \cup \neg\psi_{n+1}, & \text{else .} \end{cases}$$

Then $\Gamma^+ = \bigcup_n \Gamma_n$ is our atom. Idea: Show for each Γ_n is consistent for each n.

□

Lemma 8. *Assume $\psi_1 U \psi_2 \in A_i$. Then there is an $n \in \mathbb{N}$ with $A_i R_x A_{i+1} R_x ... R_x A_n$ s.t. $\psi_2 \in A_n$ and for all $i \leq j < n : \psi_1 \in A_j$.*

Lemma 9. *Assume the atom A_0. Then there exists an infinite fulfilling path that extends A_0, i.e. there is some path $A_0 A_1 ...$ with $A_i R_x A_{i+1}$ for all $i \in \mathbb{N}$ s.t. for any $\psi_1 U \psi_2 \in A_i$: there exists an $j \geq i$ s.t. $\psi_2 \in A_j$ and for all $i \leq k < j$ $\psi_1 \in A_k$*

Before proving that Lemma, we will prove some other useful lemmas. For that, let $\mathcal{C}$ be the set of all possible atoms. Note: this set is at most $2^{Cl(\phi)}$ large. For an atom C we denote $\hat{C} = \bigwedge_{\psi \in C} \psi$.

Lemma 10. *We can derive the following:*

$$\vdash \bigvee_{C \in \mathcal{C}} \hat{C}$$

Proof. Result from Propositional Logic. □

Lemma 11. *Let C be some atom. Then*

$$\vdash \hat{C} \to X(\bigvee_{A \in \mathcal{C}} \hat{A})$$

Proof.

$\vdash \bigvee_{A \in \mathcal{C}} \hat{A}$
$\vdash \Box \bigvee_{A \in \mathcal{C}} \hat{A}$
$\vdash \Box X \bigvee_{A \in \mathcal{C}} \hat{A}$
$\vdash \hat{C} \to X(\bigvee_{A \in \mathcal{C}} \hat{A})$ $\Box\phi \to (\psi \to \phi)$ is a tautology

□

Lemma 12. *Let C, D be some atoms such that $(C, D) \notin R_x$. Then*

$$\vdash \hat{C} \rightarrow X\neg\hat{D}$$

Proof. There are different reasons for $(C, D) \notin R_x$. First, there is a formula $X\psi \in Cl(\phi)$ s.t. $X\psi \in C$ and $\psi \notin D$, i.e., $\neg\psi \in D$.

$$\begin{array}{ll}
\vdash \hat{C} \rightarrow X\psi & \\
\vdash \psi \rightarrow \neg\neg\psi & \\
\vdash X\psi \rightarrow X\neg\neg\psi & \text{X is monotonic + 2} \\
\vdash \hat{C} \rightarrow X\neg\neg\psi & \\
\vdash \hat{D} \rightarrow \neg\psi & \\
\vdash \neg\neg\psi \rightarrow \neg\hat{D} & \\
\vdash X\neg\neg\psi \rightarrow X\neg\hat{D} & \\
\hat{C} \rightarrow X\neg\hat{D} &
\end{array}$$

Second case is $X\psi \notin A$ i.e., $\neg X \in C$ but $\psi \in D$.

$$\begin{array}{l}
\vdash \hat{C} \rightarrow \neg X\psi \\
\vdash X\psi \leftrightarrow \neg X\psi \\
\vdash \neg X\psi \rightarrow X\neg\psi \\
\vdash \hat{C} \rightarrow X\neg\psi \\
\vdash \hat{D} \rightarrow \psi \\
\vdash \neg\psi \rightarrow \neg\hat{D} \\
\vdash X\neg\psi \rightarrow X\neg\hat{D} \\
\hat{C} \rightarrow X\neg\hat{D}
\end{array}$$

Third, there is a formula $\psi_1 U \psi_2$ s.t. $\psi_1 U \psi_2, \neg\psi_2 \in A$ and $\neg(\psi_1 U \psi_2) \in B$

$$\begin{array}{l}
\vdash \hat{C} \rightarrow (\psi_1 U \psi_2 \wedge \neg\psi_2) \\
\vdash \psi_1 U \psi_2 \leftrightarrow \psi_2 \vee (\psi_1 \wedge X(\psi_1 U \psi_2)) \\
\vdash \hat{C} \rightarrow X(\psi_1 U \psi_2) \\
\vdash \hat{D} \rightarrow \neg(\psi_1 U \psi_2) \\
\vdash \neg\neg(\psi_1 U \psi_2) \rightarrow \neg\hat{B} \\
\vdash \psi_1 U \psi_2 \rightarrow \neg\neg(\psi_1 U \psi_2) \\
\vdash \psi_1 U \psi_2 \rightarrow X\neg\hat{D} \\
X(\psi_1 U \psi_2) \rightarrow X\neg\hat{D} \\
\hat{C} \rightarrow X\neg\hat{D}
\end{array}$$

Last, we have $\psi_1 U \psi_2 \in D$, $\psi_1 \in C$ and $\psi_1 U \psi_2 \notin C$, i.e., $\neg(\psi_1 U \psi_2) \in C$. From unraveling $\vdash \neg(\psi_1 U \psi_2) \rightarrow (\neg\psi_1 \vee \neg X(\psi_1 U \psi_2))$. Since $\vdash \hat{C} \rightarrow \neg X(\psi_1 U \psi_2)$ □

Definition 13. Let G_i be some global atom. We say an atom A (that is in T_a) is G_i-allowed if $G_i \vdash [a]\psi$ then $A \vdash \psi$

Lemma 14. *For any $\langle a\rangle\phi \in G_i$, there exists an infinite allowed fulfilling path $A_0A_1...$with $\phi \in A_i$*

But the idea doesnt work yet. The constructed T_a could contain not allowed atoms anymore.

Definition 15. Assume a global path $G_0G_1...$ and some $A_0A_1...$ that is allowed w.r.t. to their position. Then we define $(n, j, A_j)R_a(n', j', A_k)$ iff $A_jR_xA'_j$, $j' = j + 1$ and $n = n'$ where n and n' will be the 'ID' of the path and j,j' are the atom positions in the global path s.t. A_j or A'_j are allowed respectively.

Lemma 16. *Assume some allowed path $A_0A_1...$ in T_a w.r.t. $G_0G_1....$ Then $\forall a, b\, \forall A_i$: $\langle b\rangle\psi \in A_i \iff \langle b\rangle\psi \in G_i$.*

Proof. Assume $\langle b\rangle\psi \in G_i$.

$$\begin{array}{rcl} & \vdash & \langle a\rangle\langle b\rangle\psi \to \langle b\rangle\psi \\ & \vdash & [a](\langle a\rangle\langle b\rangle\psi \to \langle b\rangle\psi) \\ & \vdash & [a]\langle a\rangle\langle b\rangle\psi \to [a]\langle b\rangle\psi \\ G_i & \vdash & \langle a\rangle\langle b\rangle\psi \\ G_i & \vdash & [a]\langle a\rangle\langle b\rangle\psi \\ G_i & \vdash & [a]\langle b\rangle\psi \end{array}$$

By the allowed definition we conclude $A_i \vdash \langle b\rangle\psi$ and hence $\langle b\rangle\psi \in A_i$. Conversly, assume $\langle b\rangle\psi \in A_i$ and $\neg\langle b\rangle\psi \in G_i$. By axiom $\langle a\rangle\langle b\rangle\psi \leftrightarrow \langle b\rangle\psi$ we have $G_i \vdash \neg\langle b\rangle \to \neg\langle a\rangle\langle b\rangle\psi$. But the derived formula is the same as $\neg\langle b\rangle \to [a]\neg\langle b\rangle\psi$, and hence $G_i \vdash [a]\neg\langle b\rangle\psi$. By allowed definition it follows $A_i \vdash \neg\langle b\rangle\psi$ which is a contradiction. □

Lemma 17. *Assume a global path $G_0G_1....$ Then for any $\psi \in Cl(\phi)$:*

$$\psi \in G_i \text{ iff } ((G_iG_{i+1}...), (G_1...G_i)) \vDash \psi$$

For every agent a, every $A_0A_1...$ path in T_a and for any $\psi \in Cl(\phi)$:

$$\psi \in A_i \text{ iff } ((A_iA_{i+1}...), (A_1...A_i)) \vDash \psi$$

Proof. We show both on double induction on the length of ϕ. We first prove the global case. Assume $\phi = p$. We define the truth on each atom to be $L(G_i) = \{p \mid p \in G_i\}$. Then it follows:

$$p \in G_i \iff p \in L(G_i) \iff (G_i...G_\infty, G_0...G_i) \vDash p$$

. If $\phi = \neg\psi$, then we have:

$$\neg\psi \in G_i \iff \psi \notin G_i \overset{\text{IH}}{\iff} (G_i...G_\infty, G_0...G_i) \nvDash \psi \iff (G_i...G_\infty, G_0...G_i) \vDash \neg\psi$$

. $\phi = \psi_1 \vee \psi_2$ is done analogously. Assume $\phi = X\psi \in G_i$. Then $X\phi \in G_i \iff \psi \in G_{i+1} \overset{\text{IH}}{\iff} (G_{i+1}...G_\infty, G_0...G_{i+1}) \vDash \psi \iff (G_i...G_\infty, G_0...G_i) \vDash X\psi$.
Next we deal with the Until case, so assume $\phi = \psi_1 U \psi_2$. We prove two directions. If $\psi_1 U \psi_2 \in G_i$, then by construction of the global path there exists some $j \geq i$ s.t. $\psi_2 \in G_j$ and for all $i \leq k < j$: $\psi_1 \in G_k$. By IH, it follows $(G_j...G_\infty, G_0...G_j) \vDash \psi_2$ and $(G_k...G_\infty, G_0...G_k) \vDash \psi_1$. Hence, $(G_i...G_\infty, G_0...G_i) \vDash \psi_1 U \psi_2$. Now assume $(G_i...G_\infty, G_0...G_i) \vDash \psi_1 U \psi_2$. By the semantics that means there is some $j \geq i$ s.t. $(G_j...G_\infty, G_0...G_j) \vDash \psi_2$ and for all $i \leq k < j$: $(G_k...G_\infty, G_0...G_k) \vDash \psi_1$. Now IH gives us $\psi_2 \in G_j$ and hence $\psi_1 U \psi_2 \in G_j$. Because $G_{j-1} R_x G_j$, and by IH $\psi_1 \in G_{j-1}$, we conclude $\psi_1 U \psi_2 \in G_{j-1}$. Applying this argument repeatedly, we get $\psi_1 U \psi_2 \in G_i$.
Last assume $\phi = \langle a \rangle \psi$. Then by Lemma 14 there exists an infinite allowed fulfilling path $A_0 A_1 ...$ s.t. $\phi \in A_i$. By IH, that means $(A_i...A_\infty, A_0...A_i) \vDash \psi$. But that also means $(G_i...G_\infty, G_0...G_i) \vDash \langle a \rangle \psi$. Conversly, if $G_i...G_\infty, G_0...G_i \vDash \langle a \rangle \psi$, then there exists some $A_0 A_1 ...$ s.t. $(A_i...A_\infty, A_0...A_i) \vDash \psi$. Applying IH gives us $\psi \in A_i$. By construction of our T_a A_i is G_i-allowed. Assume $\neg\langle a \rangle \psi \in G_i$. Because A_i is G_i-allowed, we have $\neg\psi \in A_i$ which is a contradiction.

Next we proof the standpoint case. Here we only consider $\phi = \langle b \rangle \psi$ for some agent b. The other cases are similar. So let $\langle b \rangle \psi \in A_i$. Under the assumption that for all $\langle b \rangle \psi \in A_i \iff \langle b \rangle G_i$ we have that $\langle b \rangle \psi \in G_i$. By Lemma 14 we can construct some $B_0 B_1 ...$ s.t. $\psi \in B_i$. Applying IH gives us $(B_i...B_\infty, B_0...B_i) \vDash \psi$. Hence, $(A_i...A_\infty, A_0...A_i) \vDash \langle b \rangle \psi$. Conversly, assume $(A_i...A_\infty, A_0...A_i) \vDash \langle b \rangle \psi$. That means there exists some there is some $B_0 B_1 ... B_i$ s.t. $B_i...B_\infty, B_0...B_i \vDash \psi$. By IH, we conclude $\psi \in B_i$. By construction of T_b we know that B_i is G_i-allowed. So we can conclude $\langle b \rangle \psi \in G_i$. Assume it would not. Then $G_i \vdash \neg\langle b \rangle \psi$ and hence $G_i \vdash [b]\neg\psi$. By the definition of allowed that means $B_i \vdash \neg\psi$. But thats a contradiction. Finally, $\langle b \rangle \in A_i$ holds.

□